

Boss Challenge — Paper 21 (Class 7)

Nutrition in Plants | Lines & Angles | Winds, Storms & Cyclones | Rational Numbers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Organisms such as green plants that make their own food from simple raw materials are described as:
(A) decomposers
(B) autotrophs
(C) heterotrophs
(D) parasites
2. Two angles whose measures add up to exactly 90° are said to be:
(A) adjacent angles
(B) complementary angles
(C) vertically opposite angles
(D) supplementary angles
3. The push that air exerts on the surfaces it touches, in every direction, is called air:
(A) humidity
(B) speed
(C) pressure
(D) temperature
4. A number that can be written in the form p/q , where p and q are integers and q is not zero, is called a:
(A) irrational number
(B) rational number
(C) prime number
(D) whole number only
5. The green pigment in leaves that captures sunlight for making food is called:
(A) haemoglobin
(B) chlorophyll
(C) chlorine
(D) starch
6. Two angles whose measures add up to exactly 180° are said to be:
(A) equal angles
(B) supplementary angles
(C) complementary angles
(D) reflex angles

7. Air that is in motion from one place to another is simply called:
- (A) cloud
 - (B) pressure
 - (C) vapour
 - (D) wind
8. In a rational number written as p/q , the one value that q (the denominator) must never take is:
- (A) a large number
 - (B) zero
 - (C) a negative number
 - (D) one
9. The process by which green plants use sunlight to make their own food is called:
- (A) photosynthesis
 - (B) digestion
 - (C) respiration
 - (D) transpiration
10. The complement of an angle measuring 35° is:
- (A) 55°
 - (B) 125°
 - (C) 145°
 - (D) 65°
11. Winds always blow from a region of higher air pressure towards a region of:
- (A) higher temperature
 - (B) equal air pressure
 - (C) lower air pressure
 - (D) higher air pressure
12. Which statement is true about every integer, such as 7 or -3?
- (A) Integers are irrational numbers
 - (B) No integer is a rational number
 - (C) Only positive integers are rational
 - (D) Every integer is also a rational number
13. Besides sunlight and chlorophyll, the two raw materials a green plant needs for photosynthesis are:
- (A) oxygen and starch
 - (B) nitrogen and salts
 - (C) sugar and water
 - (D) carbon dioxide and water

14. The supplement of an angle measuring 110° is:

- (A) 70°
- (B) 250°
- (C) 90°
- (D) 80°

15. When air is heated, it expands and becomes lighter. As a result, the warm air tends to:

- (A) rise upward
- (B) stay perfectly still
- (C) sink downward
- (D) turn into water

16. Written in standard form (lowest terms with a positive denominator), the rational number $4/(-8)$ becomes:

- (A) $1/2$
- (B) $-1/2$
- (C) $-4/8$
- (D) -2

17. The gas given out by a green plant during photosynthesis in daylight is:

- (A) nitrogen
- (B) carbon dioxide
- (C) water vapour
- (D) oxygen

18. When two adjacent angles are placed so that their outer arms form a straight line, they make a linear pair, and their measures always add to:

- (A) 180°
- (B) 270°
- (C) 360°
- (D) 90°

19. When warm air rises from a place, the air pressure left behind at that place becomes:

- (A) unchanged
- (B) higher
- (C) lower
- (D) doubled

20. The additive inverse (the number you add to get 0) of the rational number $3/7$ is:

- (A) $-3/7$
- (B) 0
- (C) $7/3$
- (D) $3/7$

21. The tiny pores on the surface of a leaf through which gases enter and leave are called:

- (A) veins
- (B) stomata
- (C) roots
- (D) petals

22. When two straight lines cross, the pair of angles directly opposite each other (vertically opposite angles) are always:

- (A) always right angles
- (B) supplementary (adding to 180°)
- (C) equal in measure
- (D) complementary (adding to 90°)

23. The basic reason winds blow over the Earth is the:

- (A) light of the stars
- (B) rotation of clouds
- (C) uneven heating of the Earth by the Sun
- (D) pull of the Moon

24. The rational number $-\frac{5}{8}$ lies on the number line between the two integers:

- (A) -1 and -2
- (B) -1 and 0
- (C) 5 and 8
- (D) 0 and 1

25. Each stoma (leaf pore) is opened and closed by a pair of curved cells called:

- (A) guard cells
- (B) root hairs
- (C) blood cells
- (D) nerve cells

26. Two straight roads cross each other. One of the four angles formed measures 70° . The angle vertically opposite to it measures:

- (A) 70°
- (B) 20°
- (C) 90°
- (D) 110°

27. During the day at a coast, the land heats up faster than the sea, so the breeze blows:

- (A) straight upward only
- (B) in no particular direction
- (C) from the land towards the sea
- (D) from the sea towards the land

28. The reciprocal (multiplicative inverse) of the rational number $-\frac{2}{3}$ is:
- (A) $-\frac{3}{2}$
 - (B) $\frac{2}{3}$
 - (C) $-\frac{2}{3}$
 - (D) $\frac{3}{2}$
29. The food made by a green leaf is stored mainly in the form of:
- (A) vitamins
 - (B) starch
 - (C) fat
 - (D) protein
30. An angle that measures exactly 90° is called a:
- (A) straight angle
 - (B) right angle
 - (C) reflex angle
 - (D) acute angle
31. At night near a coast, the land cools faster than the sea, so the breeze blows:
- (A) from the land towards the sea
 - (B) there is never any breeze at night
 - (C) from the sea towards the land
 - (D) only straight downward
32. The value of $(-\frac{3}{4}) + (\frac{1}{4})$ is:
- (A) $-\frac{1}{2}$
 - (B) -1
 - (C) $-\frac{4}{4}$
 - (D) $\frac{1}{2}$
33. To test a leaf for the presence of starch, we add a few drops of:
- (A) common salt
 - (B) lime water
 - (C) blue litmus
 - (D) iodine solution
34. An angle whose measure is more than 90° but less than 180° is called:
- (A) a right angle
 - (B) an obtuse angle
 - (C) an acute angle
 - (D) a straight angle

35. If you blow hard across the top of a strip of paper held to your lips, the free end of the paper surprisingly:

- (A) is pushed downward
- (B) tears in half
- (C) rises up
- (D) stays perfectly flat

36. Which of these two negative rational numbers is the greater (lies further to the right on the number line): $-2/3$ or $-3/4$?

- (A) they are equal
- (B) $-3/4$ is greater
- (C) $-2/3$ is greater
- (D) neither can be compared

37. Plants that cannot make their own food and instead live on and take food from another living plant are called:

- (A) autotrophs
- (B) insectivores
- (C) saprotrophs
- (D) parasites

38. Two angles of a linear pair are equal to each other. The measure of each angle is:

- (A) 60°
- (B) 180°
- (C) 90°
- (D) 45°

39. Generally, the faster the air moves over a region, the air pressure there becomes:

- (A) unchanged
- (B) lower
- (C) zero
- (D) higher

40. A rational number that lies exactly between $1/4$ and $1/2$ on the number line is:

- (A) $3/8$
- (B) $1/3$
- (C) $1/8$
- (D) $5/8$

41. A pitcher plant traps and digests insects mainly to obtain the nutrient it lacks in marshy soil, namely:

- (A) oxygen
- (B) water
- (C) nitrogen
- (D) sugar

42. A transversal crosses two parallel lines. A pair of corresponding angles formed at the two crossings are always:

- (A) supplementary (adding to 180°)
- (B) always 90°
- (C) equal in measure
- (D) complementary (adding to 90°)

43. A very large, violent storm with extremely high-speed winds whirling in a spiral around a calm centre is called a:

- (A) rainbow
- (B) fog
- (C) cyclone
- (D) breeze

44. The product $(\frac{2}{3}) \times (-\frac{9}{4})$ is:

- (A) -6
- (B) $\frac{3}{2}$
- (C) $-\frac{3}{2}$
- (D) $-\frac{18}{12}$

45. Organisms such as bread mould and mushrooms that feed by breaking down dead and decaying matter are called:

- (A) herbivores
- (B) saprotrophs
- (C) autotrophs
- (D) parasites

46. A transversal cuts two parallel lines. A pair of alternate interior angles are always:

- (A) supplementary
- (B) reflex
- (C) complementary
- (D) equal in measure

47. The calm, low-pressure region right at the centre of a cyclone is called the:

- (A) tail
- (B) root
- (C) crest
- (D) eye

48. The value of $(-\frac{6}{7}) \div (-\frac{2}{7})$ is:

- (A) $\frac{1}{3}$
- (B) $\frac{3}{7}$
- (C) 3
- (D) -3

49. Fungi and algae living together so closely that both benefit, as seen in a lichen, is an example of:
- (A) competition
 - (B) parasitism
 - (C) predation
 - (D) symbiosis
50. A transversal cuts two parallel lines. A pair of co-interior (allied) angles on the same side of the transversal always add up to:
- (A) 360°
 - (B) 90°
 - (C) 180°
 - (D) they are equal
51. The instrument used to measure the speed of the wind is called an:
- (A) rain gauge
 - (B) thermometer
 - (C) anemometer
 - (D) barometer
52. Written in standard form, the rational number $(-7)/(-3)$ becomes:
- (A) $-7/-3$
 - (B) $7/3$
 - (C) $3/7$
 - (D) $-7/3$
53. The soil bacterium that lives in the root nodules of pulse (legume) plants and supplies them nitrogen is:
- (A) amoeba
 - (B) Rhizobium
 - (C) yeast
 - (D) chlorophyll
54. One angle of a linear pair measures 125° . The other angle measures:
- (A) 75°
 - (B) 45°
 - (C) 65°
 - (D) 55°
55. The very high-speed winds of a cyclone, together with the very low pressure at its centre, are dangerous mainly because they can:
- (A) light up the sky like a rainbow
 - (B) rip off roofs and uproot trees
 - (C) freeze the sea into ice
 - (D) make the weather cooler and pleasant

56. The sum of any rational number and its additive inverse is always:

- (A) 0
- (B) a negative number
- (C) 1
- (D) the number itself

57. Green plants need nitrogen to build proteins, but they cannot use the nitrogen gas of the air directly. They take nitrogen mostly as:

- (A) nitrogen gas through the leaves
- (B) starch from the roots
- (C) oxygen from the air
- (D) nitrogen salts dissolved in soil water

58. Two complementary angles are in the ratio 1 : 2. The larger of the two angles measures:

- (A) 45°
- (B) 60°
- (C) 30°
- (D) 120°

59. Towering storm clouds and thunderstorms build up most readily when the local air happens to be:

- (A) hot and humid
- (B) freezing and snowy
- (C) cool and clear
- (D) cold and dry

60. The product of any non-zero rational number and its reciprocal is always:

- (A) 0
- (B) the number itself
- (C) 1
- (D) the reciprocal itself

61. Inside a leaf cell, photosynthesis actually takes place in tiny green structures called:

- (A) cell walls
- (B) nuclei
- (C) chloroplasts
- (D) vacuoles

62. A reflex angle is an angle whose measure is:

- (A) more than 180° but less than 360°
- (B) exactly 90°
- (C) exactly 180°
- (D) less than 90°

63. Rising warm, moist air is important for storms because, as it goes up and cools, the water vapour in it:

- (A) turns into oxygen gas
- (B) disappears completely
- (C) sinks back to the ground as vapour
- (D) condenses to form clouds

64. Between any two different rational numbers on the number line, how many rational numbers can be found?

- (A) exactly one
- (B) exactly ten
- (C) none at all
- (D) infinitely many

65. If a green leaf is destarched and then kept in the dark for two days, an iodine test on it will show:

- (A) no blue-black colour, because no starch was made
- (B) a bright red colour
- (C) a strong blue-black colour everywhere
- (D) a green colour as before

66. Two supplementary angles are in the ratio 1 : 2. The two angles are:

- (A) 45° and 90°
- (B) 90° and 90°
- (C) 60° and 120°
- (D) 30° and 60°

67. When a cyclone is approaching a coast, the most sensible action for people living there is to:

- (A) climb the tallest tree
- (B) rush to the beach to watch the waves
- (C) move to a safe shelter further inland
- (D) ignore the warning and stay outdoors

68. The value of $(1/2) - (3/4)$ is:

- (A) $1/4$
- (B) $-2/4$
- (C) $5/4$
- (D) $-1/4$

69. Animals, fungi and most bacteria, which depend on plants or other organisms for their food, are called:

- (A) autotrophs
- (B) producers
- (C) pigments
- (D) heterotrophs

70. A wind vane (weathercock) on a roof first points due North, then turns clockwise to point due East. The angle it has turned through is:

- (A) 90°
- (B) 45°
- (C) 270°
- (D) 180°

71. Just before a cyclone strikes the coast, the sea water near the shore often rises sharply in a so-called storm surge. This rise mainly causes:

- (A) clearer, calmer weather
- (B) flooding of low-lying coastal land
- (C) the sea freezing over
- (D) a sudden drop in sea level

72. On a winter night a hill-station thermometer reads 2°C , then falls by 5 degrees by dawn. The dawn temperature, as a rational number of degrees, is:

- (A) 3
- (B) -3
- (C) -7
- (D) 7

73. Because a leaf takes in raw materials and uses light to make food, a leaf is often called the plant's:

- (A) waste bin
- (B) flower
- (C) food factory
- (D) water tank

74. A weathercock turns from pointing North all the way round clockwise to point South. The angle turned is:

- (A) 45°
- (B) 180°
- (C) 360°
- (D) 90°

75. A wind vane (weathercock) mounted on a roof is used to show the:

- (A) speed of the wind in km/h
- (B) amount of rainfall
- (C) temperature of the air
- (D) direction from which the wind is blowing

76. During a storm the air pressure drops steadily, falling $\frac{1}{4}$ unit each hour for 3 hours. The total change in pressure, as a rational number, is:

- (A) $-\frac{3}{4}$
- (B) -12
- (C) $\frac{3}{4}$
- (D) $-\frac{1}{4}$

77. The very first food (a simple sugar) formed when a leaf carries out photosynthesis is:
- (A) protein
 - (B) starch
 - (C) glucose
 - (D) fat
78. Two angles are both equal AND supplementary to each other. Each angle therefore measures:
- (A) 90°
 - (B) 60°
 - (C) 180°
 - (D) 45°
79. The reason a hot-air balloon rises into the sky is that the air inside it is heated, so it becomes:
- (A) lighter than the cooler air outside
 - (B) the same weight as the outside air
 - (C) heavier than the cooler air outside
 - (D) turned into water
80. Which of these correctly shows the rational number $5/(-6)$ written with a positive denominator?
- (A) $5/6$
 - (B) $-6/5$
 - (C) $6/5$
 - (D) $-5/6$
81. Leaves look green to our eyes mainly because chlorophyll:
- (A) turns water green
 - (B) adds green colour to sunlight
 - (C) absorbs only green light
 - (D) reflects green light while absorbing other colours
82. The complement of an angle of 1° is:
- (A) 99°
 - (B) 179°
 - (C) 91°
 - (D) 89°
83. The seasonal winds that bring most of India's rain, blowing in from the sea in summer, are the:
- (A) dry desert winds
 - (B) monsoon winds
 - (C) cyclone eye winds
 - (D) land breezes
84. The product $(-2/5) \times 0$ is:
- (A) 0
 - (B) 1
 - (C) $2/5$
 - (D) $-2/5$

85. Growing pulses (legumes) in a field between crops helps the soil because their roots, with Rhizobium, restore the soil's:

- (A) sunlight
- (B) carbon dioxide
- (C) water
- (D) nitrogen

86. The sum of all the angles formed around a single point (a complete turn) is:

- (A) 360°
- (B) 180°
- (C) 90°
- (D) 270°

87. A flag on a tall pole flutters and streams out straight in one direction. This best shows that there is a strong:

- (A) rainfall over the pole
- (B) rise in air temperature
- (C) wind blowing in that direction
- (D) drop in humidity

88. Which list shows three rational numbers that all lie between 0 and 1?

- (A) $\frac{1}{2}$, 1, $\frac{3}{2}$
- (B) $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$
- (C) $-\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$
- (D) $\frac{3}{4}$, $\frac{5}{4}$, $\frac{7}{4}$

89. The water and dissolved minerals a plant uses for nutrition are absorbed from the soil by its:

- (A) fruits
- (B) roots
- (C) flowers
- (D) stomata

90. A transversal cuts two parallel lines and one co-interior (allied) angle measures 70° . Its co-interior partner on the same side measures:

- (A) 110°
- (B) 90°
- (C) 70°
- (D) 20°

91. A dark, funnel-shaped cloud that reaches down from a thundercloud to the ground, with very fast whirling winds, is called a:

- (A) rainbow
- (B) glacier
- (C) waterfall
- (D) tornado

92. Is the number 0 a rational number, and why?

- (A) Yes, because 0 can be written as $0/1$
- (B) Yes, but only when written as $1/0$
- (C) No, because 0 cannot be a fraction
- (D) No, because its denominator must be 0

93. The mode of nutrition seen in fungi growing on a rotting log, where they secrete juices to digest the dead wood and then absorb it, is:

- (A) saprotrophic nutrition
- (B) autotrophic nutrition
- (C) insectivorous nutrition
- (D) parasitic nutrition

94. An angle whose measure is less than 90° is called:

- (A) a right angle
- (B) a reflex angle
- (C) an obtuse angle
- (D) an acute angle

95. Wind speed is commonly reported by weather offices in units of:

- (A) degrees Celsius
- (B) kilograms
- (C) kilometres per hour
- (D) millimetres

96. The rational number midway between -1 and 0 on the number line is:

- (A) -2
- (B) $1/2$
- (C) -1
- (D) $-1/2$

97. The simple sugar made in photosynthesis is carried from the leaves to all other parts of the plant by tubes called:

- (A) chlorophyll
- (B) phloem
- (C) xylem
- (D) stomata

98. The supplement of an angle is three times the angle itself. The original angle measures:

- (A) 60°
- (B) 135°
- (C) 45°
- (D) 90°

99. When cooler, denser air moves in to replace warm air that has risen, this horizontal movement of air is felt by us as a:

- (A) pressure
- (B) cloud
- (C) wind
- (D) rainfall

100. A diver descends $\frac{3}{5}$ of a metre each second below the water surface. After 4 seconds her depth, as a rational number of metres relative to the surface, is:

- (A) $\frac{12}{5}$
- (B) $-\frac{3}{5}$
- (C) $-\frac{7}{5}$
- (D) $-\frac{12}{5}$